

# Test & QA Report

ArchViz Pro — AI Digital Twin Platform

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Date: 28 July 2026

Build: main @ ca0d9d5 · Version 1.0.0

Verdict: PASS (1 recommended fix)

## Executive Summary

The application is healthy and production-shippable. Every automated check that can be run in the current environment passes: TypeScript type-checking, the production build, 15 server endpoint behaviors, 9 core-logic unit checks, and a real browser render with full interaction. One security issue is worth fixing before a server-side Gemini key is configured, alongside a few minor items and one structural gap (no automated test suite / CI).

**Overall verdict: PASS** — with one recommended fix (open-proxy on the legacy AI routes).

## Test Results

| Suite                             | Result          | Notes                                            |
|-----------------------------------|-----------------|--------------------------------------------------|
| TypeScript (tsc --noEmit)         | PASS — 0 errors | Clean.                                           |
| Production build (Vite + esbuild) | PASS            | Client + server.cjs emitted.                     |
| Lint (npm run lint)               | PASS            | Lint maps to tsc.                                |
| Server smoke — unconfigured       | PASS 9/9        | 503 degradation; health flags false.             |
| Server smoke — configured         | PASS 6/6        | 401 unauth; 400 on bad webhook signature.        |
| Core logic units (BOM, geometry)  | PASS 9/9        | Grouping, cost parse, collision, room lookup.    |
| Browser render + interaction      | PASS            | Mounts; tutorial / 2D toggle / legal modal work. |

Two initial smoke “failures” were traced to a test-harness JSON-escaping bug; the endpoints were re-verified to return a correct 401 with valid request bodies.

## Build Metrics (gzip)

| Chunk    | Size (gzip) | Chunk        | Size (gzip) |
|----------|-------------|--------------|-------------|
| main app | 118.5 kB    | genai (lazy) | 61.3 kB     |
| three.js | 130.7 kB    | react        | 4.2 kB      |
| supabase | 55.9 kB     | CSS          | 9.8 kB      |

Vendor libraries are code-split and the AI SDK is lazy-loaded. The three.js chunk trips Vite’s >500 kB (raw) warning — expected for a 3D engine and already isolated into its own chunk.

## Findings by Severity

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| MEDIUM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Legacy AI proxy is an open, unmetered endpoint |
| POST /api/digitize-blueprint and /api/rebuild-from-prompt build the Gemini client with a fallback to the server GEMINI_API_KEY, and these two routes have no authentication. If the app is deployed with a server-side Gemini key, anyone who finds these URLs can consume your Gemini quota unauthenticated and unmetered. The newer /api/ai/* routes are correctly auth-gated and metered; the client no longer calls the legacy ones. Fix: require the x-gemini-api-key header on the legacy routes (no env fallback), delete them, or add auth + rate-limiting. |                                                |
| LOW                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Malformed JSON returns HTML, not JSON          |
| Express’s default body-parser error returns an HTML “Bad Request” page rather than a JSON error envelope. Fix: add a small JSON error-handling middleware.                                                                                                                                                                                                                                                                                                                                                                                                          |                                                |
| LOW                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Fonts loaded from an external CDN              |

src/index.css imports fonts from fonts.googleapis.com (the source of the single console error observed in the sandbox, where the CDN is blocked). Minor reliability/privacy nit. Fix: self-host the fonts to remove the third-party request.

**LOW****No @types/react**

JSX is loosely typed, so type-checking cannot catch prop/JSX mistakes. Fix: add @types/react and @types/react-dom and resolve the surfaced any usages in a dedicated pass.

**GAP****No automated test suite or CI**

There are no test files and no vitest/jest/CI configuration; all verification is manual. For a product now handling auth, billing and row-level security, this is the largest structural risk. Recommendation: add vitest for the pure utilities (bom, geometry, layout, watermark, plan gating) and a GitHub Actions workflow running tsc + build + tests on PRs.

## Not Verified Here (needs a staging environment)

- **Stripe** end-to-end (Checkout → webhook → subscriptions row). Prices exist; server env not yet set.
- **Supabase RLS** at runtime (tenant isolation, team permissions). SQL is written defensively; run one live cross-tenant test.
- **Hosted AI** with a real Gemini key (metering and cache-hit path).

## Strengths

- Clean type-check and build; no runtime console errors originating from app code.
- Sound security fundamentals: secrets kept server-side, JWT verification, workspace-scoped RLS, signature-verified webhooks, owner-only billing, Free-plan watermarking.
- Graceful degradation: features return 503 / empty states rather than crashing when a migration or env var is missing.
- Well-factored shared utilities and a consistent, branded UI that renders and interacts correctly.

## Recommendation

Ship-ready for a controlled launch. Before enabling a server-side Gemini key, close the MEDIUM open-proxy finding. As fast-follows: add a vitest suite plus GitHub Actions CI, self-host fonts, and add @types/react.

ArchViz Pro — AI Digital Twin Platform · Automated QA verification report. This report reflects checks executed in a sandbox without live third-party credentials; items under “Not Verified” require a staging environment.